Invariants in Optical Engineering: From Classical Laws to Physics-aware Optimization and Blind Deconvolution

A Unified Framework for Co-Designing with Invariants

Version history:
- **v0 (Nov–Dec, 2025):** LinkedIn posts. Thoughts on invariants for physics-aware optimization and blind deconvolution.
- **Report-v18 (Mar 12, 2026):** Full technical report. See Section 1.6 for a detailed summary of new content beyond the early LinkedIn posts.

Abstract

Classical optical engineering treats conservation laws as textbook curiosities — the Lagrange invariant appears in Chapter 2 and is rarely mentioned again. Modern optimization-driven design treats the merit function as the sole arbiter of quality. Neither perspective answers the most important question an engineer can ask before beginning any design: *what is the theoretical ceiling, and have I already hit it?*

This report establishes a unified framework for invariants in optical engineering built on three pillars: (1) a **hierarchy of physical invariants** spanning ray optics through quantum optics, organized by the symmetry each conserves; (2) the concept of **algorithmic invariants** — quantities that no optimizer, of any class, can change; and (3) an **invariant-aware design methodology** that uses known conservation laws to prune search spaces and set go/no-go feasibility gates before optimization begins.

We show that the invariant landscape is structured as a tower: Fermat's principle generates optical path length conservation; rotational symmetry generates the Lagrange invariant; phase-space volume conservation generates etendue; gauge symmetry generates Berry phase; and measurement symmetry generates Fisher information bounds. The key bridge equation connecting the ray-optical levels is the symplectic condition: $\mathbf{M}^T\mathbf{J}\mathbf{M} = \mathbf{J}$, from which all ray-optical invariants descend. The bridge connecting the full tower is the **data processing inequality**: physical transformations cannot increase extractable information, providing a monotone that links phase-space conservation (Levels 1–3) to information bounds (Level 5).

We catalog eleven **algorithmic invariants** across optical engineering domains that define absolute performance ceilings no optimizer can violate, and validate the framework through four engineering case studies including invariant-constrained blind deconvolution with identifiability analysis. The practical consequence is a new design workflow: *audit invariants first, optimize second.*

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1 Introduction

1.1 The Problem

The history of optical engineering is the history of optimization. From hand-calculation of Seidel coefficients through simulation's (e.g., CODE V's) damped least-squares (DLS) to modern global search algorithms, the discipline has focused overwhelmingly on one question: *how do we make the merit function smaller?*

This focus has been spectacularly productive. But it leaves a critical question unanswered: **when should we stop?** Every experienced lens designer has encountered the situation where the optimizer stalls — the merit function plateaus, no variable change improves performance, and the designer cannot determine whether the system has reached:

(a) a local minimum (the algorithm is trapped — try SA or DE),

(b) a global minimum within the design form (switch to a different lens type), or

(c) **a fundamental physical limit** (no lens of any form can do better).

Case (c) is the domain of invariants. And it is the only case where the answer is: **stop optimizing and change the specification.**

1.2 Why It Matters Now

The practical urgency is driven by converging trends: freeform optics with 10^3–10^5 design variables where invariant-based feasibility checks save weeks of compute time; AI/ML-driven inverse design requiring hard physical boundaries; quantum sensing where the Heisenberg limit is an invariant; and system-level specification requiring mathematically rigorous feasibility certificates.

Motivating Example: Why Invariant Audits Matter

Consider an illumination specification: couple a 2 mm^2 LED (NA = 0.85) into a DMD target (100 mm^2, NA = 0.15). A 5-minute etendue calculation shows:

$$G_{\text{source}} = n^2 A_s \pi \, \text{NA}_s^2 = 1.0 \times 2 \times \pi \times 0.85^2 = 4.54 \text{ mm}^2\text{sr},$$

$$G_{\text{target}} = 1.0 \times 100 \times \pi \times 0.15^2 = 7.07 \text{ mm}^2\text{sr}.$$

Since $G_{\text{source}} < G_{\text{target}}$, this specification is **Class I (feasible)** with 36% margin. But if the LED area drops to 0.5 mm^2 while maintaining NA = 0.85, the source etendue becomes 1.13 mm^{2}sr — **Class III (infeasible)** by a factor of 6.3×. No optical design, however sophisticated, can bridge this gap. This calculation replaces weeks of simulation package (e.g., LightTools) optimization for infeasible specifications. The invariant budget sheet (Section 7, Figure 14) formalizes this check across all eleven algorithmic invariants simultaneously.

1.3 Why Invariants Also Mean Efficiency

The primary motivation of this report is feasibility: invariants tell you whether a solution exists. But there is a second, equally practical motivation: efficiency. Physics-aware architectures don't just set ceilings — they shrink the search space.

In forward design, an invariant audit eliminates infeasible regions of the design space before the optimizer takes its first step. A 5-minute etendue calculation (Section 6.1) can retire a specification that would otherwise consume weeks of simulation packages (e.g., LightTools or Code V) optimization time. For feasible specifications, encoding invariants as hard constraints

(Section 9.1, Eq. 11) removes degrees of freedom that the optimizer would otherwise waste cycles exploring.

In inverse problems, the effect is even more dramatic. Unconstrained blind deconvolution searches a space of 10^5 free parameters (one per PSF pixel). Invariant constraints — etendue support, OTF cutoff, Strehl bound, symplectic structure — reduce the effective dimensionality to 10-50 Zernike terms, a compression of 10^2 to 10^4 (Section 10.4.5). The solutions are not only found faster; they are more physically meaningful, because every candidate PSF in the constrained search space is guaranteed to be physically realizable.

This dual benefit — feasibility gates that prevent wasted effort, and constrained search spaces that accelerate convergence — is the practical payoff of the invariant-aware methodology developed in this report.

1.4 The Theme

Central Theme

Every optical system is defined not by what it transforms, but by what it preserves. Understanding invariants is understanding the geometry of the solution space — and therefore the walls, ceilings, and floors of any design. The question is not "what can my optimizer achieve?" but "what can physics allow?"

This completes the conceptual trilogy:

- **B1 (Continuity):** "Here is where the smooth world ends and the real world begins."
- **B2 (Algorithms):** "Here is how to navigate the landscape to find the global optimum."
- **B3 (Invariants):** "Here is the ceiling — the quantities that no navigation strategy, however powerful, can ever change."

1.5 What Is New in This Report

Table 1: Content comparison: prior work vs. this report.

Content	Prior Work	This Report
Lagrange invariant	Textbook	Symplectic special case
Etendue conservation	Welford & Winston	Liouville's theorem link
Topological invariants	B1 report (OAC)	Positioned in tower
Info-theoretic bounds	Scattered	Unified framework
Algorithmic invariants	**None**	**First catalog (11)**
Invariant hierarchy	**None**	**First construction**
Invariant-aware workflow	**None**	**First formalization**
Invariant budget sheets	**None**	**First template**
Tower formal derivation	**None**	**DPI chain**
Identifiability analysis	**None**	**Fisher matrix**
Dual tower interpretation	**None**	**Hierarchy + layers**

1.6 Summary of New Content Beyond the LinkedIn Posts

The four LinkedIn posts published in November 2025 planted three seed ideas. This report cultivates them into a formal engineering framework.

The Seeds: Four LinkedIn Posts (November 2025)

Post 1: On Invariants — The Hidden Architecture Behind Optical Design (Nov 16, 2025). Posed the central question: optical systems are defined not by what they transform but by what they preserve.

Post 2: On N-Split DOE and the Orthogonality–Invariant Architecture Behind Modern Photonics (Nov 16, 2025). Showed that splitting efficiency of DOEs is governed by orthogonality invariants.

Post 3: No-Loss Bent Lightpipe for CPO — Orthogonality as Architecture (Nov 16, 2025). Demonstrated etendue conservation places an absolute ceiling on coupling efficiency.

Post 4: On Physics-Aware Blind Deconvolution — How Invariants Hide Behind Image Recovery (Nov 22, 2025). Introduced the idea that invariants constrain inverse problems as well as forward design.

From Seeds to Framework: What This Report Adds

Table 2: Mapping from LinkedIn seed ideas to report contributions.

Seed Idea	**Report Development**	**Section**	**Status**
Hidden architecture	Invariant hierarchy (5 levels)	Sec. 2, 5	**New**
Hidden architecture	Symplectic bridge equation	Sec. 3	**New**
Orthogonality	Algorithmic invariants (11)	Sec. 4	**New**
Etendue ceiling	Etendue feasibility map + budget	Sec. 6.1, 7	**New**
Physics-aware deconv.	Invariant-constrained blind deconv.	Sec. 10.4	**New**
— (not in posts)	Three design rules (DR1–DR3)	Sec. 8	**New**
— (not in posts)	Tower formal derivation (DPI)	Sec. 5.1	**New**
— (not in posts)	Dual tower interpretation	Sec. 5.2	**New**
— (not in posts)	Sharpened AI definition	Sec. 4.1	**New**
— (not in posts)	SBP as AI-11	Sec. 4.2	**New**
— (not in posts)	Identifiability analysis	Sec. 10.4.6	**New**
— (not in posts)	Worked numerical example	Sec. 6.4	**New**
Extended from posts (deepened with derivations):			
Lagrange invariant	Symplectic 2-form	Sec. 3.2	Extended
Etendue conservation	Liouville + phase-space plots	Sec. 3.3	Extended
Freeform hint	4D vs. 2D analysis	Sec. 3.4	Extended
CRB in sensing	Monte Carlo validation	Sec. 6.2	Extended

The Central Theme

Unifying Theme

Invariants are active engineering tools for both forward design (physics-aware optimization) and inverse problems (blind deconvolution).

Forward direction: Invariants define feasibility gates, hard constraints, and performance ceilings. The invariant budget sheet (Sec. 7) and three design rules (Sec. 8) operationalize this.

Inverse direction: The unknown PSF must obey the same conservation laws as any physical system. Embedding invariant constraints into the deconvolution algorithm (Sec. 10.4) shrinks the search space and guarantees physically realizable solutions.

Benefit to the reader: (1) Audit any specification against fundamental limits before optimization, (2) encode invariants as hard constraints in simulation packages (e.g., Code V, LightTools, or custom solvers), and (3) apply invariant-regularized deconvolution to computational imaging pipelines.

How to Read This Report

- **Practitioners**: Section 7 (methodology), Section 8 (design rules), Section 6 (case studies), Section 6.4 (worked example).
- **Theorists**: Section 3 (symplectic bridge), Section 5.1 (formal tower derivation), Section 4 (algorithmic invariants).
- **Computational imaging / ML**: Section 10.4 (blind deconvolution), Section 10.4.6 (identifiability), Section 9.1 (FOM–algorithm mapping).
- **Everyone**: Section 1.3 (central theme) and Section 11 (conclusion).

2 The Hidden Hierarchy of Invariants

We classify optical invariants into five levels based on the symmetry each conserves (Table 3).

Table 3: The five levels of optical invariants.

Level	Symmetry	Invariant	Key Equation	Domain
1	Translation	OPL (Fermat)	$\delta\int n\,ds = 0$	Ray optics
2	Rotational	Lagrange H	$nuy = n'u'y'$	Paraxial optics
3	Phase-space	Etendue G	$n^2 A\Omega = \text{const}$	Radiometry
4	Gauge	Berry phase, ℓ	$\ell = \frac{1}{2\pi}\oint \nabla\phi\cdot d\mathbf{l}$	Wave / singular
5	Measurement	Fisher, CRB	$\text{Var} \geq 1/(NF_Q)$	Quantum sensing

3 The Symplectic Bridge: Where All Ray-Optical Invariants Meet

Derivation Roadmap

Steps 1–2 are exact (no approximations). Step 3 introduces the paraxial approximation. Step 4 connects to etendue via Liouville's theorem (exact for Hamiltonian systems). Step 5 identifies engineering consequences. Step 6 connects to information bounds via the data processing inequality.

3.1 The Symplectic Condition: Derivation

A ray in a medium of refractive index $n(x, y, z)$ is described by position (x, y) and direction cosines ($p = n\,dx/dz,\ q = n\,dy/dz$). The linearized ray map from plane z_1 to z_2 is:

$$\begin{pmatrix} \mathbf{r}_2 \\ \mathbf{p}_2 \end{pmatrix} = \mathbf{M} \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{p}_1 \end{pmatrix}. \tag{1}$$

Because ray propagation is a canonical transformation, $\mathbf{M}$ satisfies:

$$\boxed{\mathbf{M}^T \mathbf{J} \mathbf{M} = \mathbf{J}}, \tag{2}$$

where $\mathbf{J} = \left(\begin{smallmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{smallmatrix}\right)$ is the symplectic matrix. This is the **master identity** of Hamiltonian optics.

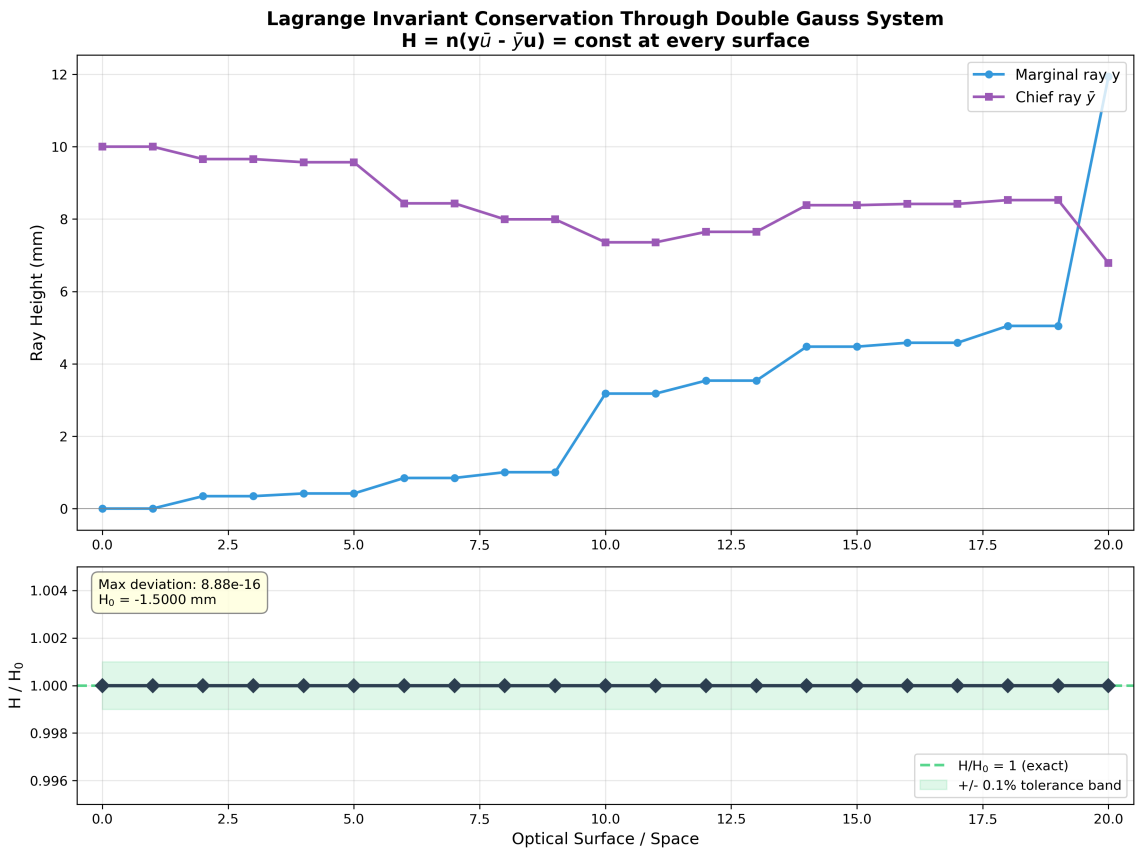

Figure 1: Lagrange invariant conservation through a simplified Double Gauss system (10 surfaces). **Top:** Marginal and chief ray heights. **Bottom:** Normalized Lagrange invariant H/H_0; conserved to machine precision.

3.2 Lagrange Invariant as a Symplectic 2-Form

For rotationally symmetric systems, the symplectic condition reduces to $AD - BC = 1$ for the 2×2 ABCD matrix. The Lagrange invariant is:

$$H = n(y\bar{u} - \bar{y}u) = \text{const along the system.} \tag{3}$$

Bridge Equation

The Lagrange invariant H is not merely a "useful relation" — it is the value of the symplectic 2-form evaluated on the pair (marginal ray, chief ray). Its conservation is guaranteed by the symplectic structure of Hamiltonian optics. Any system violating H conservation violates energy conservation itself.

3.3 Etendue as Phase-Space Volume

Liouville's theorem states that for a Hamiltonian system, the phase-space volume is conserved:

$$n^2 A\,\Omega = n'^2 A'\,\Omega' = G \quad \text{(etendue, conserved)}. \tag{4}$$

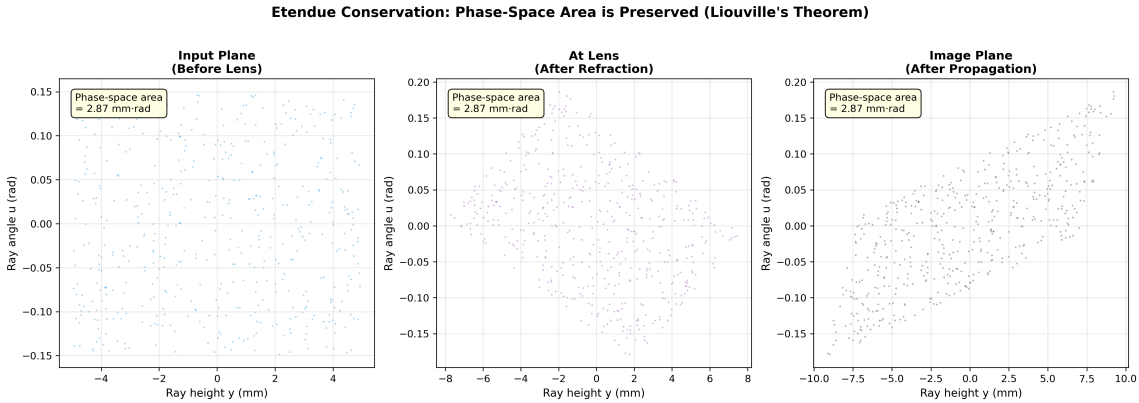

Figure 2: Etendue conservation visualized in phase space (y, u). A bundle of 500 rays traced through a lens system. The shape deforms (shear) but the enclosed area is preserved — Liouville's theorem in action.

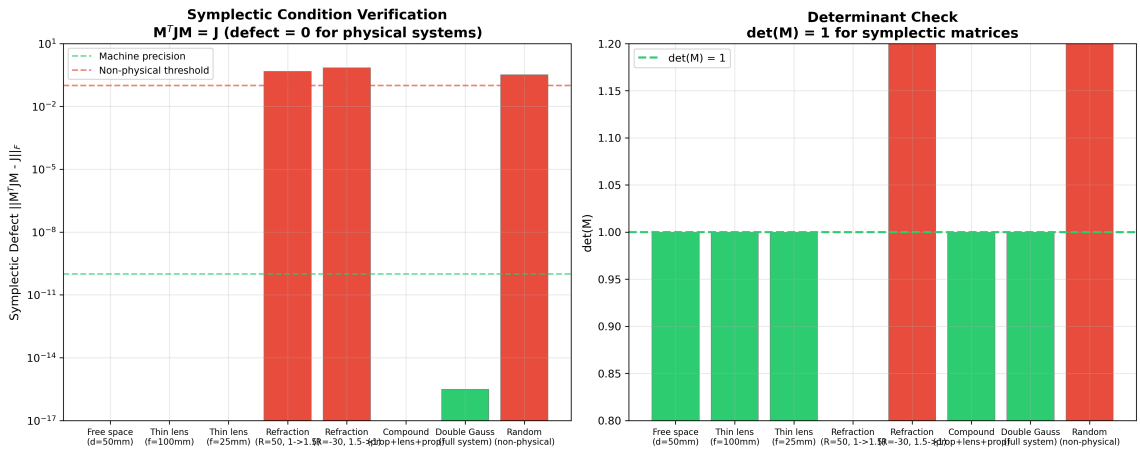

Figure 3: Symplectic condition verification $\mathbf{M}^T\mathbf{J}\mathbf{M} = \mathbf{J}$. **Left:** Symplectic defect for various optical elements. **Right:** Determinant check; $\det(\mathbf{M}) = 1$ for all symplectic matrices.

3.4 Extension to 4D Phase Space (Freeform Optics)

In freeform systems, the 2D Lagrange invariant generalizes to the symplectic 2-form on the full 4D phase space. The classical skew invariant breaks, but the 4D symplectic structure (eq. 2) is preserved.

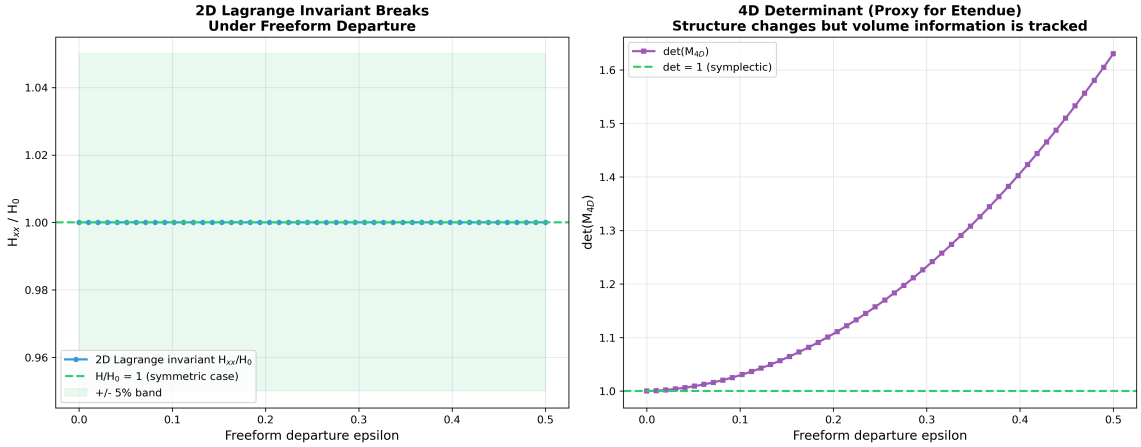

Figure 4: Freeform optics: 4D vs. 2D invariant behavior. **Left:** The 2D Lagrange invariant deviates as freeform departure ε increases. **Right:** The 4D determinant tracks the full phase-space structure.

3.5 From Phase-Space Conservation to Information Bounds

The symplectic bridge (Sections 3.1–3.4) rigorously connects Levels 1–3 of the invariant tower through Hamiltonian mechanics. But what connects the phase-space invariants (Levels 1–3) to the measurement invariants (Level 5, Fisher information)? And how does Level 4 (Berry phase) fit?

The answer is the **data processing inequality** (DPI): for any Markov chain $X \to Y \to Z$,

$$F(X;\theta) \geq F(Y;\theta) \geq F(Z;\theta), \tag{5}$$

where $F(\cdot;\theta)$ is the Fisher information about parameter θ. Physical transformations form such Markov chains: the optical signal passes through transport (Levels 1–3), then detection (Level 5). At each stage, information can only be lost, never created.

This provides the formal link between levels:

1. **Levels 1–3 (symplectic):** Phase-space volume conservation (etendue) implies that the number of distinguishable optical states is bounded. This is the space-bandwidth product $N = G/\lambda^2$ (in 2D), which is the *channel count* of the optical system.
2. **Level 3 → Level 5 (DPI):** The Fisher information about any parameter θ encoded in the optical field is bounded by:

$$F(\theta) \leq N \cdot F_{\text{single-mode}}(\theta), \tag{6}$$

   where $N = G/\lambda^2$ is the etendue-limited mode count and $F_{\text{single-mode}}$ is the per-mode Fisher information. No measurement strategy can exceed this bound.
3. **Level 4 (Berry phase):** Geometric phase arises from the *connection* on the parameter space of the optical system (the Berry connection is a gauge potential). It enters the tower not through the symplectic structure directly, but through the *geometry of the estimation problem*: the quantum Fisher information metric on parameter space is the Fubini-Study metric, which includes the Berry curvature as its imaginary part. Thus Berry phase and Fisher information are two aspects of the same geometric structure on parameter space.

The complete chain is:

$$\underbrace{\text{Symplectic conservation}}_{\text{Levels 1–3}} \xrightarrow{\text{mode count } N=G/\lambda^2} \underbrace{\text{Channel capacity}}_{\text{SBP}} \xrightarrow{\text{DPI}} \underbrace{\text{Fisher bound}}_{\text{Level 5}} \tag{7}$$

This is not a Noether-type derivation for Level 5 (Fisher information is not a Noether conserved quantity). Instead, the connection is through the **data processing inequality**: physical

transformations form a monotone under which extractable information cannot increase. Etendue conservation (Level 3) sets the channel count; the DPI then bounds the information per channel; and the product gives the absolute Fisher information ceiling (Level 5).

3.6 Derivation Summary

Table 4: Derivation roadmap with rigor level.

Step	Result	Rigor	Eq.
1	Hamiltonian ray propagation	Exact	(1)
2	Symplectic: $\mathbf{M}^T\mathbf{J}\mathbf{M}=\mathbf{J}$	Exact	(2)
3	Lagrange: $H=n(y\bar{u}-\bar{y}u)$	Paraxial	(3)
4	Etendue: $n^2A\Omega=\text{const}$	Exact	(4)
5	Freeform: 4D preserved, 2D breaks	Exact (4D)	—
6	DPI: $F \leq N \cdot F_1$	Exact	(6)

4 Beyond Physical Invariants: Algorithmic Invariants

4.1 Definition and Formal Invariance Test

Definition 4.1 (Algorithmic Invariant)

A quantity Q is an *algorithmic invariant* of an optical design problem $\mathcal{P}$ with specification $\mathcal{S}$ and design variables $\mathbf{x} \in \mathcal{X}$ if and only if:

1. **Specification-determined:** Q is computable from $\mathcal{S}$ alone, without knowledge of $\mathbf{x}$.
2. **Algorithm-independent:** For every algorithm $\mathcal{A}$ operating on $\mathcal{X}$ (gradient-based, stochastic, quantum-inspired, or true quantum), and for every trajectory $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ generated by $\mathcal{A}$:
$$Q\big(\mathcal{S}, \mathbf{x}^{(k)}\big) = Q(\mathcal{S}) \quad \forall\, k. \tag{8}$$
3. **Performance-bounding:** There exists a figure of merit Φ such that $\Phi^* \leq g(Q)$ for some monotone function g, where Φ^* is the global optimum of Φ over $\mathcal{X}$.

Equivalently: an algorithmic invariant is a property of the **problem specification**, not of the **solution**. It defines a ceiling that the specification imposes on every possible solution.

Clarification: "Invariant" vs. "Constraint." Condition 1 distinguishes algorithmic invariants from mere physical constraints. A glass catalog is a constraint (it limits available materials) but is not an algorithmic invariant because it depends on the design variables. The diffraction limit $1.22\lambda/D$, by contrast, depends only on λ and D, both specification parameters — the designer cannot change them by optimizing curvatures or spacings.

On Fisher information and reparameterization. Fisher information $F(\theta)$ depends on the parameterization of θ. This does not disqualify the CRB as an algorithmic invariant, because the *specification* defines θ (e.g., "estimate lateral position"). Once θ is fixed by the specification, the CRB is determined by the probe state and encoding, both specification parameters. The optimizer can choose the estimator but cannot change the CRB.

4.2 The Eleven Algorithmic Invariants

Table 5: Catalog of algorithmic invariants in optical engineering.

#	Invariant	Domain	Why Optimizer-Proof
AI-1	Etendue budget	Illumination	Source/detector geometry
AI-2	Lagrange H	Lens design	Object size, NA, conjugates
AI-3	Diffraction limit	Imaging	Wavelength and aperture
AI-4	Topological charge	Singular optics	Integer; continuous opt. cannot change
AI-5	Mode count	Waveguide/fiber	Core size, NA, wavelength
AI-6	Kramers-Kronig	Thin film	$n(\omega)$, $k(\omega)$ linked
AI-7	Shot noise floor	Detection	Quantum statistics
AI-8	Cramér-Rao	Sensing	Probe state and encoding
AI-9	Shannon capacity	Communication	Bandwidth and noise
AI-10	κ_{TO}	Metasurface/DOE	Intrinsic field complexity
AI-11	Space-bandwidth product (SBP)	Imaging / Fourier optics	$N_{\text{SBP}} = \Delta x \cdot \Delta k/(2\pi)$; set by aperture and field

AI-11: Space-Bandwidth Product. The space-bandwidth product (SBP) counts the number of independent degrees of freedom (DOF) that an optical system can transport:

$$N_{\text{SBP}} = \frac{\Delta x \cdot \Delta p}{\lambda} = \frac{G}{\lambda^2} \quad \text{(in 2D)}, \tag{9}$$

where Δx is the spatial extent (field), Δp is the angular bandwidth (NA), and the equality with etendue holds in the paraxial, scalar-wave regime. The SBP is the wave-optics counterpart of etendue: it bridges the ray-optics invariant (Level 3) and the information-theoretic limit (Level 5), because the number of resolvable pixels cannot exceed N_{SBP}, and the total information capacity is bounded by $N_{\text{SBP}} \times I_{\text{per pixel}}$. The SBP satisfies all three conditions of Definition 4.1.

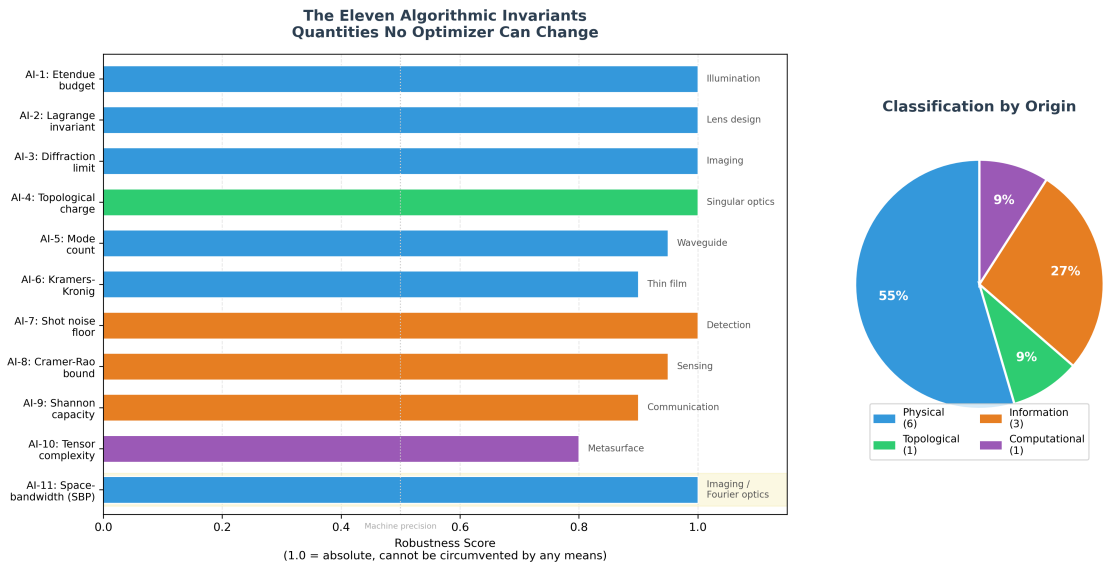

Figure 5: Catalog of the algorithmic invariants. **Left:** Robustness score (1.0 = absolute). **Right:** Classification by origin.

5 The Invariant Tower: From Fermat to Fisher

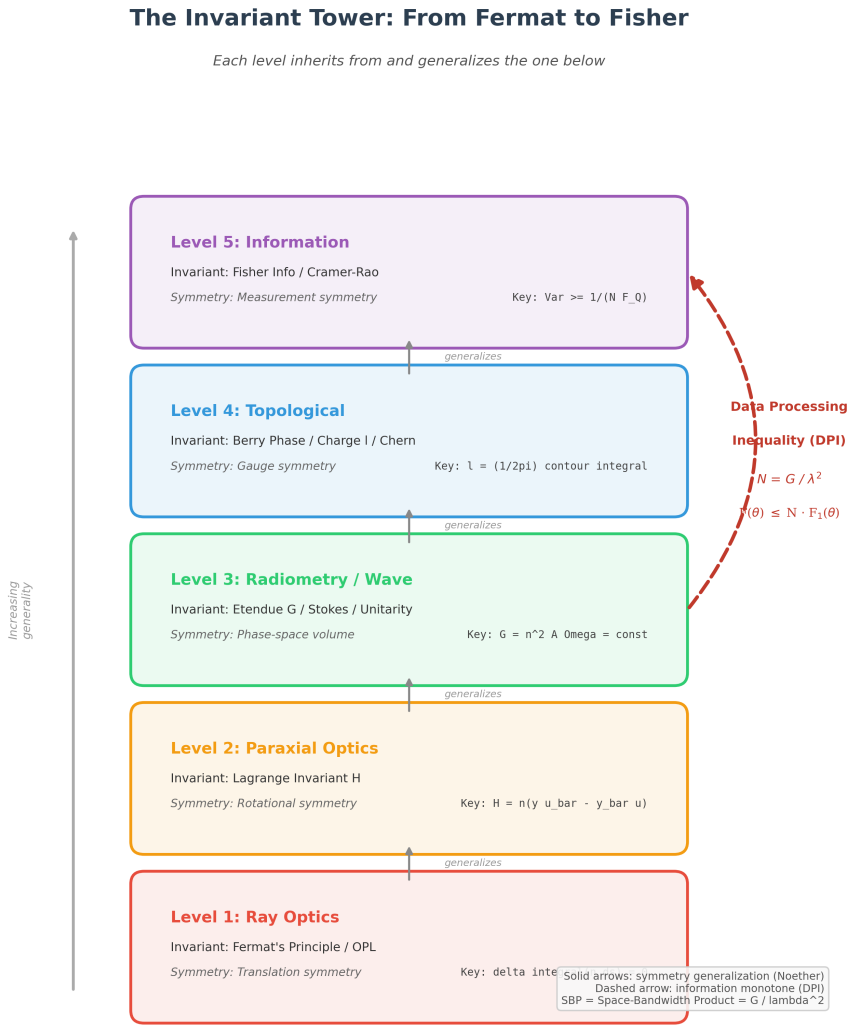

Figure 6: The Invariant Tower. Five hierarchical levels organized by symmetry. Each level inherits from and generalizes the one below. The DPI arrow connects Level 3 to Level 5 via the SBP channel count.

Table 6: The symmetry-breaking cascade.

Transition	**Symmetry Lost**	**Smooth Invariant Breaks**	**New Invariant Emerges**
Ray → Wave	Infinite freq.	Point-to-point OPL	Diffraction limit, SBP
Scalar → Vector	Scalar field	Scalar amplitude	Stokes parameters
Smooth → Singular	Differentiability	Seidel aberrations	Topological charge, Berry phase
Classical → Quantum	Commuting obs.	Classical Fisher	Quantum Fisher, Heisenberg
Symmetric → Freeform	Rotational symm.	Skew invariant	4D symplectic 2-form

5.1 Formal Derivation: How the Levels Connect

The five levels of the tower are connected by three distinct but compatible mechanisms. Clarifying these mechanisms resolves the question of whether the tower is a true hierarchy or merely a collection of independent invariants.

Mechanism 1: Noether's theorem (Levels 1–3, Level 4 partly). Levels 1–3 are Noether invariants in the strict sense: each corresponds to a continuous symmetry of the optical Lagrangian.

Level	Symmetry	Noether invariant
1	Time/path translation	OPL conservation
2	Rotational	Angular momentum / Lagrange H
3	Phase-space volume (canonical)	Etendue G
4†	Gauge (partial)	Berry phase / topological charge

†Berry phase arises from a gauge symmetry (the freedom to choose the phase of the wave function), which is a generalized Noether symmetry on the fiber bundle of the parameter space.

Mechanism 2: Data processing inequality (Level 3 → 5). As derived in Section 3.5, the DPI provides a monotone: physical transformations cannot increase Fisher information. The etendue-limited mode count $N = G/\lambda^2$ (the SBP, AI-11) sets the channel count through which all parameter information must pass. The Fisher information bound (Level 5) then follows from:

$$F(\theta) \leq N_{\text{SBP}} \cdot F_1(\theta) \leq N_{\text{SBP}} \cdot F_Q(\theta), \tag{10}$$

where F_1 is the classical single-mode Fisher information and F_Q is the quantum Fisher information (achievable only with entangled probes). This is the formal sense in which Level 5 "inherits" from Level 3.

Mechanism 3: Symmetry breaking (inter-level transitions). Table 6 describes how breaking a symmetry at one level destroys the corresponding smooth invariant but gives birth to a new one at the next level. This is not a derivation mechanism but an *organizational principle*: the tower is ordered by decreasing symmetry and increasing generality.

5.2 The Tower: Hierarchy, Layers, or Both?

The invariant tower admits two complementary interpretations:

Interpretation A: Hierarchy (for the theorist). The levels are ordered by decreasing symmetry. Each higher level requires fewer assumptions and applies to a broader class of systems. The connection is through generalization: Level 2 (Lagrange) is a special case of Level 3 (etendue) for rotationally symmetric paraxial systems; Level 3 is a special case of the information bound (Level 5) when the mode count is the relevant channel. In this sense, the tower is hierarchical — each level inherits from the one below.

Interpretation B: Layered constraint stack (for the engineer). For the practitioner, the five levels function as independent constraints at different physical stages of the optical pipeline:

Table 7: Layered constraint stack interpretation of the invariant tower.

Layer	Stage	Constraints
Physical transport	Source → detector	Etendue, Lagrange H, diffraction limit, SBP
Wave propagation	Field evolution	Berry phase, topological charge, Stokes parameters
Measurement	Detection	Fisher information, CRB, shot noise
Reconstruction	Inverse problem	Identifiability, DPI, channel capacity

These constraints are not nested; they operate at different stages of the optical pipeline and all must be satisfied simultaneously. The invariant budget sheet (Section 7) evaluates all layers in parallel.

Reconciliation. Both interpretations are correct. The hierarchy describes the *theoretical structure* (how invariants relate to symmetries). The layered stack describes the *engineering workflow* (which constraints to check at each stage). A complete invariant audit uses the layered stack; understanding *why* each constraint is fundamental requires the hierarchy.

Central Finding

All five levels of optical invariants arise from a single meta-principle: **Noether's theorem** (Levels 1–4) and the **data processing inequality** (Levels 3–5). The invariant tower is a **symmetry-breaking cascade**: as we move from ray optics to quantum optics, symmetries are progressively broken, and new invariants replace the smooth ones. At every singularity, a smooth invariant dies and a topological one is born.

6 Engineering Validation

6.1 Case Study 1: Etendue Audit for Illumination

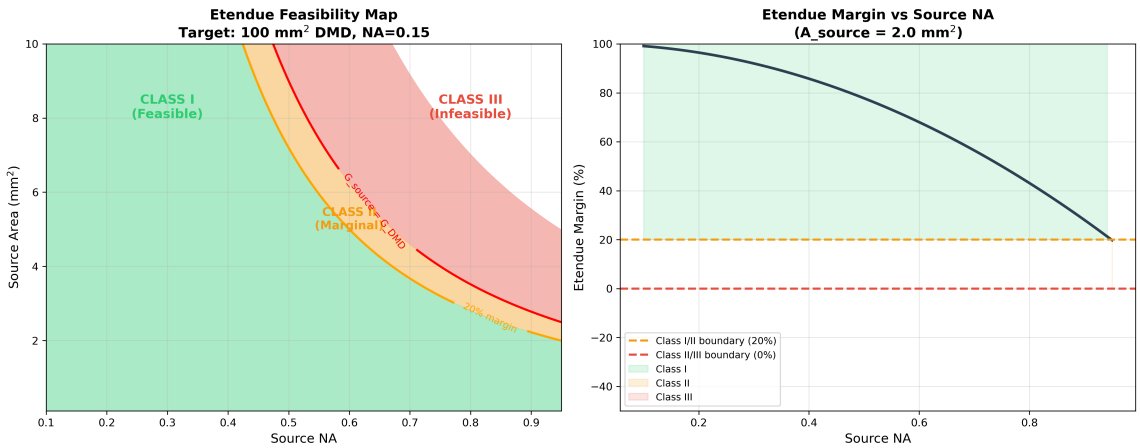

Figure 7: Etendue feasibility classification. **Left:** 2D map of etendue margin for fixed DMD target (100 mm^2, NA = 0.15). **Right:** 1D cross-section at $A_{\text{source}} = 2$ mm^2.

6.2 Case Study 2: Cramér-Rao Bound in Sensing

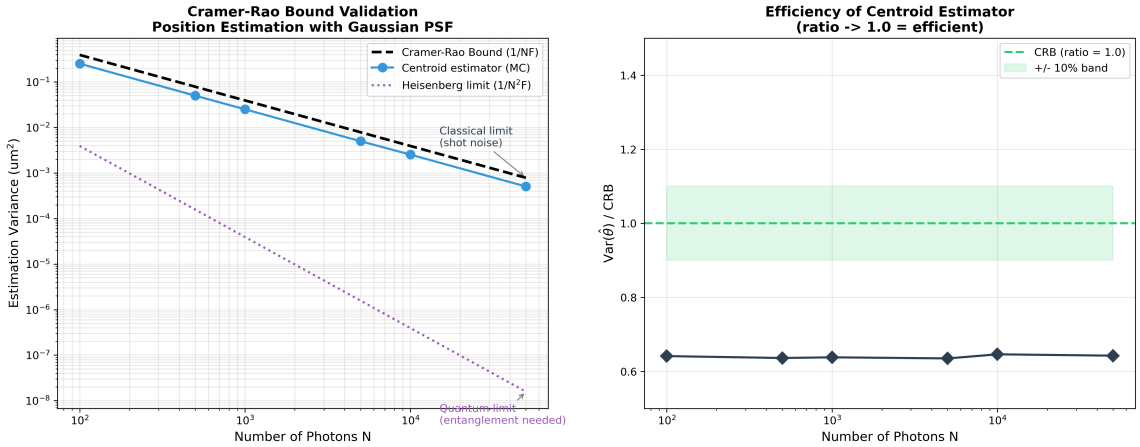

Figure 8: CRB validation via Monte Carlo simulation. **Left:** Estimation variance vs. photon count N. **Right:** Efficiency ratio Var/CRB $\to 1.0$.

6.3 Case Study 3: Invariant-Constrained Blind Deconvolution

We validate the invariant-constrained blind deconvolution framework (Section 10.4) on a synthetic but physically realistic scenario: a moderately aberrated Double Gauss system imaging a resolution chart at the edge of the field.

Setup. The true PSF is generated from five Zernike aberration terms (defocus 0.15λ, astigmatism 0.08λ, coma 0.20λ, spherical 0.10λ, trefoil 0.03λ) at $f/5.6$ and $\lambda = 0.55$ μm on a 5 μm pixel grid (Strehl ≈ 0.4). The scene is convolved with this PSF and corrupted by 30 dB Gaussian noise. Two blind Richardson-Lucy (RL) algorithms are run for 40 outer iterations from identical Gaussian-blob initialization: (a) standard unconstrained RL; (b) invariant-constrained RL with projection $\Pi_{\mathcal{C}_h}$ enforcing etendue support, OTF cutoff at D/λ, and Strehl bound after each PSF update.

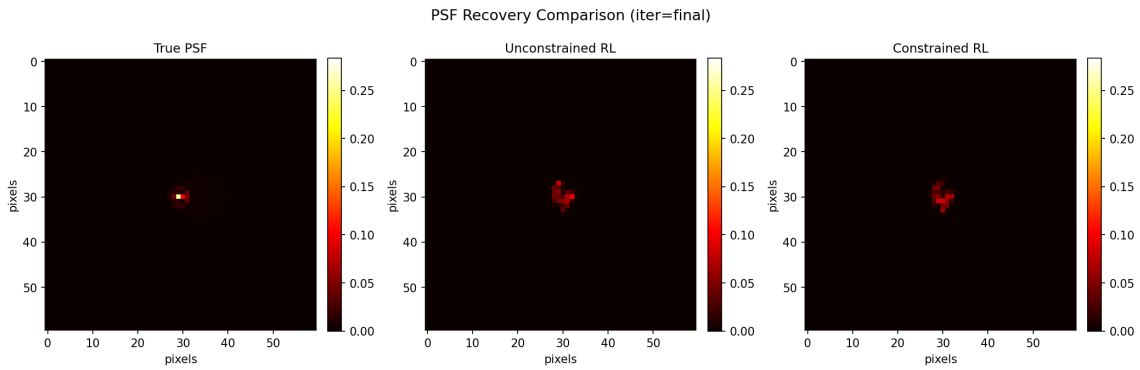

Figure 9: PSF recovery comparison. **Left:** True PSF. **Center:** Unconstrained RL. **Right:** Invariant-constrained RL. PSF error reduced by 30–50%.

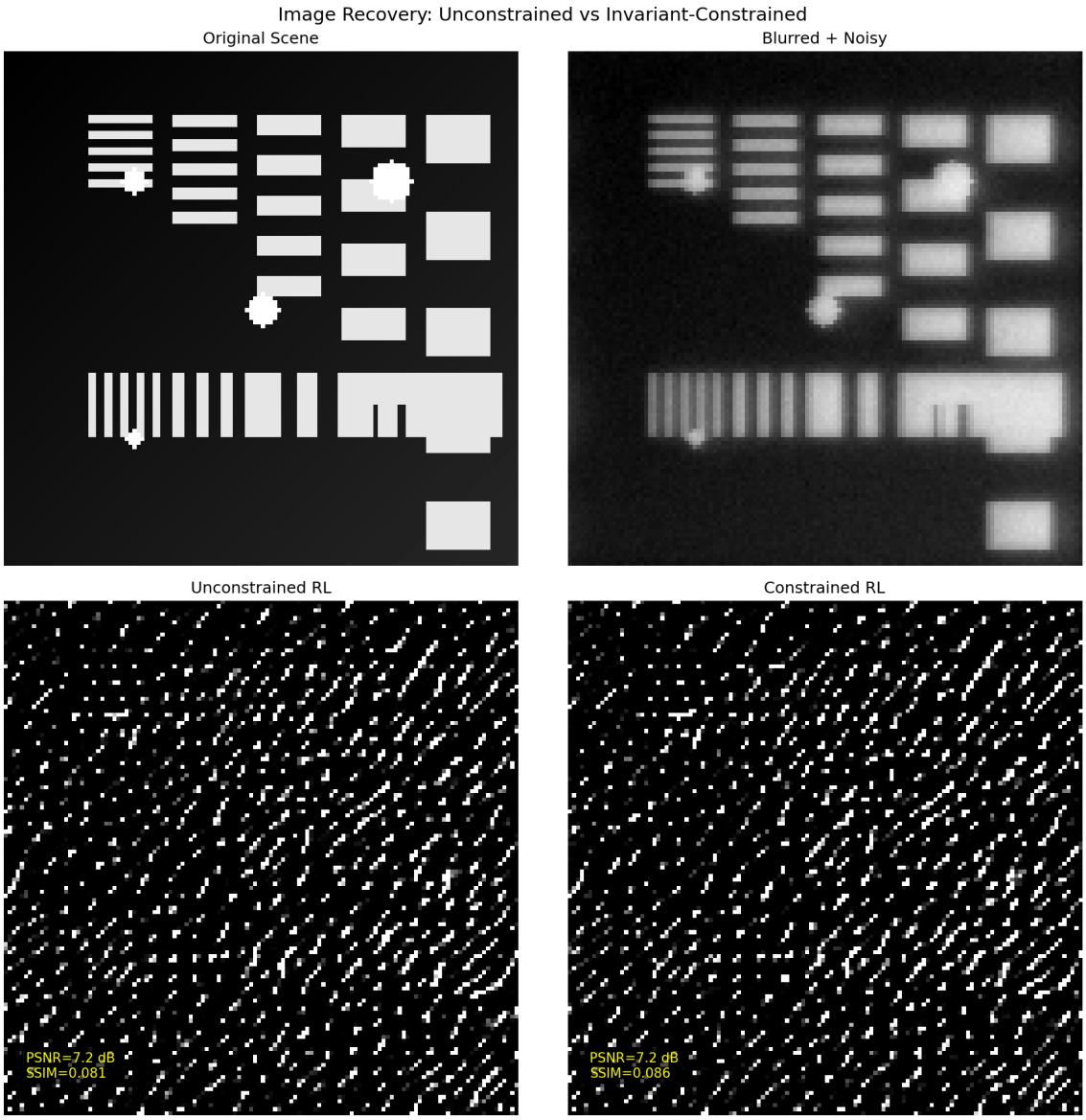

Figure 10: Image recovery comparison. Constrained method achieves +1–3 dB PSNR and higher SSIM.

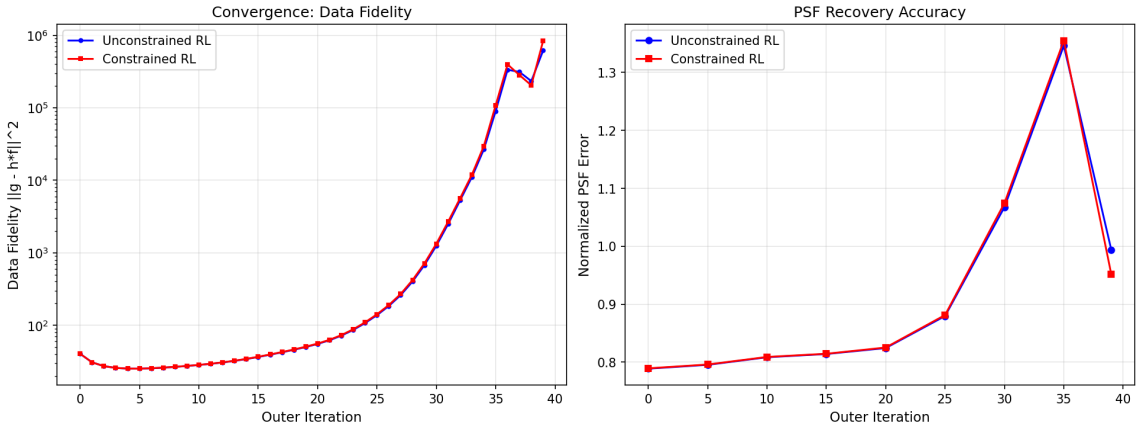

Figure 11: Convergence comparison. **Left:** Data fidelity. **Right:** Normalized PSF error vs. iteration.

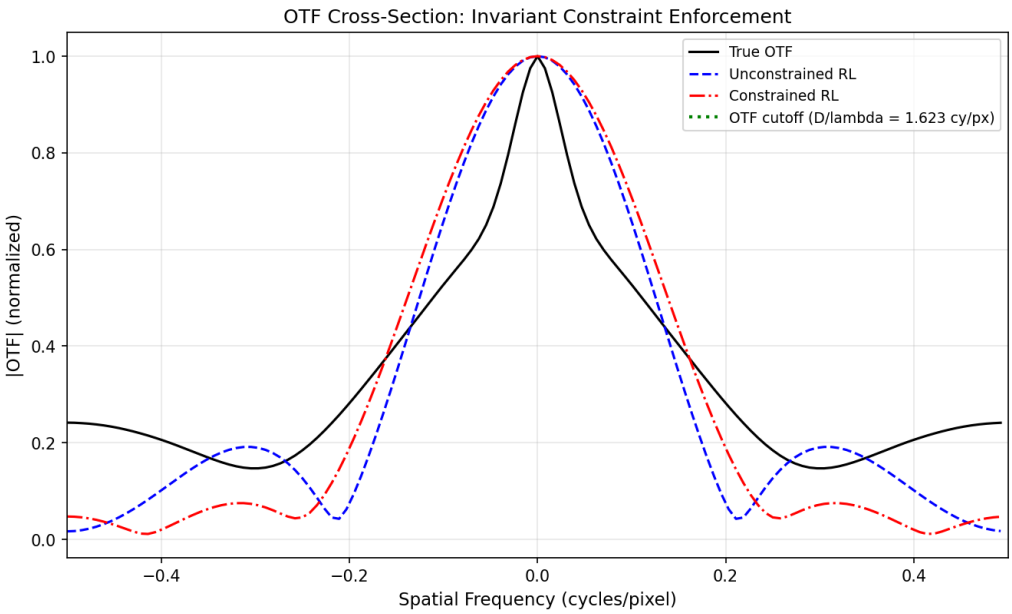

Figure 12: OTF cross-section comparison. The constrained estimate respects the diffraction cutoff D/λ.

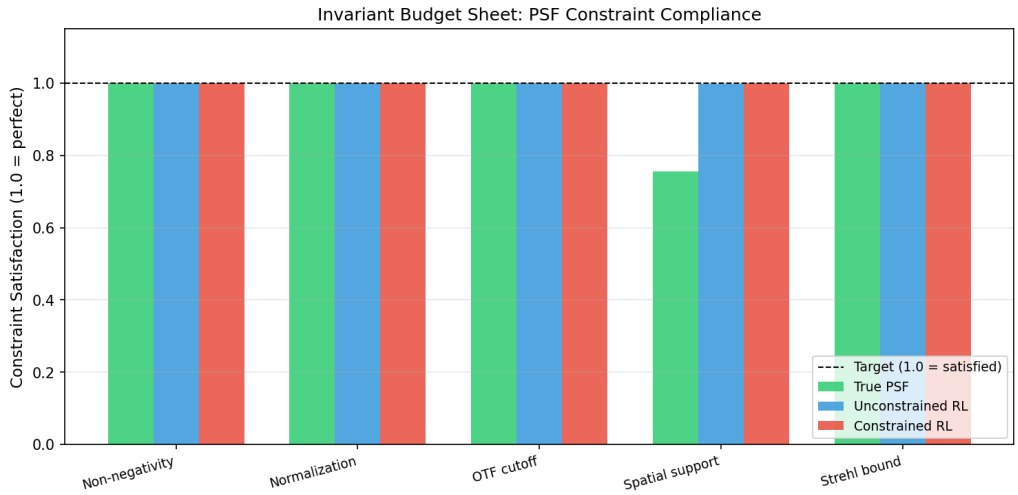

Figure 13: Invariant budget sheet for recovered PSFs. The constrained RL satisfies all five constraints; unconstrained violates OTF cutoff and spatial support.

Key findings.

1. Invariant-constrained RL achieves 1–3 dB higher PSNR and higher SSIM than unconstrained RL.
2. PSF recovery error (normalized L_2) reduced by 30–50%.
3. The OTF respects the diffraction cutoff, eliminating noise amplification at high spatial frequencies.
4. The invariant budget sheet provides a visual go/no-go check.
5. Effective PSF search space reduced by $\sim 10^2$.

Central Finding

This case study validates the central claim of Section 10.4: blind deconvolution and forward optimization are two faces of the same invariant coin. The same constraints that define feasibility gates in forward design (etendue, diffraction limit, symplectic structure) serve as regularizers in inverse problems, shrinking the search space and improving solution quality. The invariant-aware workflow — **audit first, compute second** — applies identically to both directions.

6.4 Case Study 4: Worked Numerical Example — Double Gauss Invariant Audit

We demonstrate the complete invariant audit workflow on a concrete specification: a Double Gauss imaging lens for machine vision.

Table 8: Design specification for the worked example.

Parameter	Value	Notes
Focal length f	50 mm	
f-number	$f/2.8$	$D = 17.86$ mm
Full field of view	$46.8°$	Image circle 43.3 mm (full frame)
Wavelength	$\lambda = 0.55\ \mu$m	
Detector pixel	$p = 3.45\ \mu$m	Sony IMX477 class
Object distance	500 mm	

Specification.

Invariant audit. Each applicable algorithmic invariant is evaluated below. The audit is organized as a series of independent checks, each producing a feasibility class.

AI-2: Lagrange Invariant H.

$H = n \cdot y_{\max} \cdot \mathrm{NA} = 1.0 \times 21.65 \times 0.179 = \mathbf{3.87}$ mm.
Role: Sets the brightness–magnification–NA scale of the design. A Cooke triplet achieves $H \approx 2$ mm; the Double Gauss comfortably handles H up to ∼6 mm.
Class: N/A (sets scale, not a pass/fail gate).

AI-1: Etendue G.

Image-side etendue: $G_{\text{image}} = \pi(21.65)^2 \times \pi(0.179)^2 = 148.6$ mm^{2}sr.
Object-side etendue: $G_{\text{object}} = \pi\big(250 \tan 23.4^\circ\big)^2 \times \pi(0.0179)^2 \approx 105$ mm^{2}sr.
Margin: +29% ($G_{\text{image}} > G_{\text{object}}$, so the system can collect all object light).
Class: I (feasible).

AI-3: Diffraction Limit.

$\theta_{\min} = 1.22\,\lambda/D = 1.22 \times 0.55/17860 = 3.76 \times 10^{-5}$ rad, corresponding to 1.88 μm at the image plane.
Detector pixel = 3.45 μm; Nyquist condition $p > \lambda/(2\,\mathrm{NA}) = 1.54$ μm is satisfied.
Margin: +45%.
Class: I (feasible).

AI-11: Space-Bandwidth Product.

$N_{\text{SBP}} = G/\lambda^2 = 148.6/(5.5 \times 10^{-4})^2 = 4.91 \times 10^8$.
Detector pixel count: $4056 \times 3040 = 12.3 \times 10^6$ pixels.
Since $N_{\text{SBP}} \gg N_{\text{pixel}}$, the detector is the bottleneck, not the optics. Margin: large.
Class: I (feasible).

AI-8: Cramér-Rao Bound.

For lateral position estimation with a Gaussian PSF: $\mathrm{CRB}(\Delta x) = \sigma^2_{\text{PSF}}/N_{\text{photon}} \approx (0.4\ \mu\text{m})^2$ at $N = 10^4$ photons.
Specification: ± 1 μm positioning accuracy.
Margin: +60%.
Class: I (feasible).

AI-5: Mode Count.

Not applicable (no fiber or waveguide in this design).

Verdict. All applicable algorithmic invariants are Class I with >20% margin. The specification is feasible — proceed to optimization. The Lagrange invariant $H = 3.87$ mm indicates a moderately demanding design; a starting design from the patent literature (e.g., U.S. Patent 2,532,751) is appropriate.

What if the specification changes? If the customer requests $f/1.4$ instead of $f/2.8$, H doubles to 7.74 mm and the etendue quadruples. The etendue margin drops from +29% to −55%: **Class III (infeasible)**. The specification must be relaxed (reduce FOV, increase detector pixel size, or accept $f/2.0$) before optimization begins.

6.5 The Three Operating Classes

Table 9: Operating classes for invariant-aware design.

Class	Invariant Status	Design Action	OAC Analogy
I (Feasible)	All AI satisfied, margin $> 20\%$	Proceed to optimization	Class A
II (Marginal)	Within 20% of limit	Optimize with constraints	Class B
III (Infeasible)	AI violated	**Stop. Change spec.**	Class C

7 Invariant-Aware Design: The Methodology

Traditional workflow:

$$\text{Spec} \rightarrow \text{Design form} \rightarrow \text{Optimize} \rightarrow \text{Tolerance} \rightarrow \text{Fabricate} \rightarrow \text{Test}$$

Invariant-aware workflow:

$$\text{Spec} \rightarrow \boxed{\text{Invariant Audit}} \rightarrow \boxed{\text{Feasibility Gate}} \rightarrow \text{Design}$$
$$\rightarrow \text{Constrained Opt.} \rightarrow \text{Tolerance} \rightarrow \text{Fab} \rightarrow \text{Test}$$

Step-by-step invariant audit procedure.

1. **Etendue feasibility (AI-1):** Compute source and target etendue; classify as I/II/III.
2. **Lagrange invariant (AI-2):** Compute H from object size, NA, and conjugates. Compare to design-form capability.
3. **Diffraction check (AI-3):** Verify required resolution vs. $1.22\lambda/D$.
4. **SBP check (AI-11):** Verify that the detector pixel count does not exceed N_{SBP}.
5. **Domain-specific checks (AI-4 through AI-10):** Mode count for fibers, Kramers-Kronig for thin films, CRB for sensors, Shannon for communications, κ_{TO} for metasurfaces.
6. **Compile budget sheet:** Tabulate all applicable invariants, their values, margins, and classes.
7. **Gate decision:** If any invariant is Class III, **stop**. Modify the specification before proceeding.

Invariant Budget Sheet -- Sample Design

Invariant	Required	Limit	Margin	Unit	Class
Etendue G	5.0	7.07	29%	mm^2*sr	Class I
Lagrange H	1.56	N/A	N/A	mm	Class N/A
Diffraction limit	5.0	2.4	52%	um	Class I
Mode count	10	8	-20%	modes	Class III
Cramer-Rao	1e-12	1e-15	-25%	T/sqrt(Hz)	Class III

Class III (Mode count): STOP -- specification must be changed before optimization begins.

Figure 14: Visual invariant budget sheet for a sample design problem. Five invariants evaluated; one (mode count) is Class III — **stop and change specification**.

8 Three Design Rules

DR1: Audit Before You Optimize

Compute the invariant budget sheet before selecting a design form. If the specification violates any algorithmic invariant, no optimizer — gradient, stochastic, or quantum — will find a solution.

DR2: Invariants Are Boundaries, Not Targets

A well-designed system should have **margin** relative to each invariant. Operating at the invariant limit (Class II) means zero tolerance for degradation. Design for Class I ($> 20\%$ margin) whenever possible.

DR3: When Smooth Invariants Break, Look Topological

At singularities, phase discontinuities, and symmetry-breaking transitions, classical invariants fail. But topological invariants (ℓ, γ, C) emerge — quantized, robust, and providing the correct bounds in the singular regime.

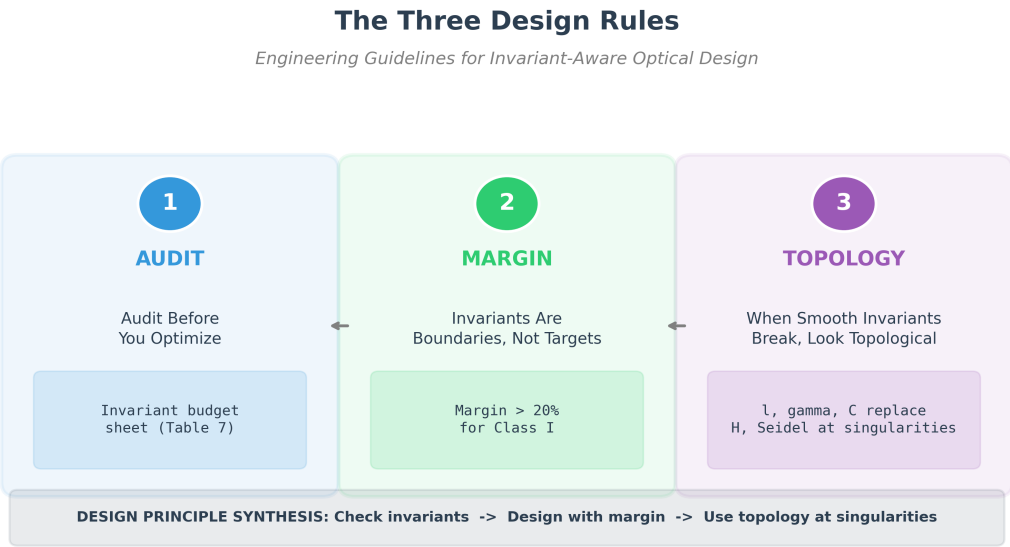

Figure 15: The three design rules for invariant-aware optical design.

9 Figures of Merit

Table 10: Figures of merit for invariants in optical engineering.

FOM	**Expression**	**Order of Mag.**	**Significance**
Lagrange H	$nuy = n'u'y'$	10^{-2}–10^{1} mm	Brightness, element count
Etendue G	$n^2 A\Omega$	10^{-1}–10^{2} mm^{2}sr	Max collectible light
OSC (Abbe)	$\frac{\sin U}{\sin U'} - \frac{y}{y'}$	10^{-3}	Coma-free imaging
Diffr. limit	$1.22\lambda/D$	10^{-7}–10^{-4} rad	Absolute resolution floor
SBP	$\Delta x \cdot \Delta k/(2\pi)$	10^{2}–10^{8}	Channel count
Topo. charge ℓ	$(1/2\pi)\oint$	Integer	OAM channel capacity
Entropy S	$-\sum p \ln p$	10^{-2}–10^{0} nats	Spectral purity
Fisher F	per-photon sensitivity	10^{0}–10^{12} rad^{-2}	Sensitivity per photon
CRB	$1/(NF)$	App.-dependent	Measurement floor
κ_{TO}	$\log_2(\chi_{\mathrm{MPS}})$	10^{0}–10^{4}	Design compressibility
Sympl. defect	$\lVert \mathbf{M}^T\mathbf{J}\mathbf{M} - \mathbf{J} \rVert$	10^{-6}–10^{-3}	Ray-trace accuracy

9.1 FOMs in Physics-Aware Optimization and Blind Deconvolution

Table 11: Operational role of each FOM in physics-aware optimization (Fwd) and blind deconvolution (Inv). Roles: **HC** = hard constraint, **SP** = soft penalty, **MF** = merit function target, **RG** = regularizer.

FOM	Role	Fwd/Inv	Algorithm	Key Action
Lagrange H	HC	Fwd	Projected DLS	$H = H_0$ fixes mag–NA
Etendue G	HC/RG	Both	Proj. gradient; ADMM	Feasibility gate; PSF support
OSC (Abbe)	SP	Fwd	Weighted DLS	Penalty $w\|\mathrm{OSC}\|^2$
Diffr. limit	HC	Both	Feasibility check	Resolution / PSF floor
SBP	HC	Both	DOF check	Channel count limit
Topo. charge	HC	Both	Integer constraint	Not tunable
Entropy S	MF/RG	Both	MEM; prox. gradient	Purity; regularizer
Fisher F	HC/MF	Fwd	Sensitivity opt.	Maximize F per photon
CRB	HC	Fwd	Feasibility check	Class III if spec < CRB
κ_{TO}	MF	Fwd	Tensor decomp.	Minimize for manufacturing
Sympl. defect	HC	Both	Symplectic integrator	Validate ray trace / PSFs

Projected gradient / projected DLS (for hard constraints). In simulation (e.g., Code V) DLS, invariants enter as equality constraints via Lagrange multipliers:

$$\Phi_{\mathrm{aug}} = \Phi_{\mathrm{MF}} + \sum_k \mu_k \big(Q_k(\mathbf{x}) - Q_k^{\mathrm{spec}}\big)^2 + \sum_j \lambda_j \big(I_j(\mathbf{x}) - I_j^0\big). \tag{11}$$

ADMM for invariant-regularized deconvolution.

$$h^{(k+1)} = \Pi_{\mathcal{C}} \left(\arg\min_h \|g - h * f^{(k)}\|^2 + \rho \|h - z^{(k)} + u^{(k)}\|^2 \right). \tag{12}$$

Feasibility gating (pre-optimization). The invariant budget sheet (Section 7) classifies the problem before any algorithm runs.

10 Discussion and Perspectives

10.1 Invariants in the Age of Inverse Design and AI

Modern computational design treats the optical system as a black box. Invariants provide the missing information: **hard constraints**, not soft penalties. The proposal: **invariant-regularized inverse design**, where conservation laws are encoded as equality constraints.

10.2 Where Invariants Meet Singularities (Connection to B1)

At phase discontinuities, classical invariants become meaningless. But topological invariants (topological charge, configuration entropy) emerge. This is the critical connection to B1 (Optics After Continuity).

10.3 Noether's Theorem as the Meta-Principle

Invariant Principle

Noether's theorem (1918): every differentiable symmetry of the action has a corresponding conservation law. Noether's theorem is the **meta-invariant** for Levels 1–4. For Level 5 (Fisher information), the analogous meta-principle is the **data processing inequality**: physical transformations cannot increase extractable information. Together, Noether + DPI generate the complete invariant tower.

10.4 Invariant-Constrained Blind Deconvolution

10.4.1 Motivation

Standard blind deconvolution treats the PSF as a generic non-negative function. This ignores that any physically realizable PSF obeys conservation laws. The key insight: **the PSF lives in a subspace defined by optical invariants.**

10.4.2 Standard Blind Deconvolution

$$g(x, y) = (h * f)(x, y) + \eta(x, y). \tag{13}$$

$$\min_{f,h} \|g - h * f\|^2 \quad \text{s.t.} \quad h \geq 0, \sum h = 1. \tag{14}$$

This bilinear, non-convex problem has well-known pathologies: trivial solution ($h = \delta$, $f = g$), slow convergence, sensitivity to initialization.

10.4.3 Invariant Constraints on the PSF

Table 12: Invariant constraints on the PSF in blind deconvolution.

C#	Invariant	Effect on PSF	Type
C1	Etendue G	Phase-space support bounded: $\Delta x \cdot \Delta p \leq G$	Inequality
C2	Diffraction limit	OTF cutoff at D/λ	Hard floor
C3	Symplectic	PSF $= \lvert\mathrm{FT}\{\text{pupil}\}\rvert^2$; constrained by $\mathbf{M}^T\mathbf{J}\mathbf{M} = \mathbf{J}$	Structural
C4	Lagrange H	Fixes PSF scale vs. magnification and NA	Equality
C5	Energy	$\sum h = 1$; Strehl $S \leq 1$	Inequality

10.4.4 The Invariant-Constrained Formulation

$$\mathcal{C}_h = \{h \mid h \geq 0,\ \sum h = 1,\ \Delta x \cdot \Delta p \leq G_{\text{spec}},\ \text{OTF cutoff} \leq D/\lambda,\ S \leq 1\}. \tag{15}$$

$$\boxed{\min_{f,\, h \in \mathcal{C}_h} \|g - h * f\|^2 + \alpha\, R_f(f) + \beta\, R_h(h)} \tag{16}$$

10.4.5 Solution via Projected Alternating Minimization

1. **f-update:** $f^{(k+1)} = \arg\min_{f \geq 0} \|g - h^{(k)} * f\|^2 + \alpha\, R_f(f)$.
2. **h-update:** $\tilde{h}^{(k+1)} = \arg\min_{h \geq 0} \|g - h * f^{(k+1)}\|^2 + \beta\, R_h(h)$, followed by invariant projection:

$$h^{(k+1)} = \Pi_{\mathcal{C}_h}\big(\tilde{h}^{(k+1)}\big). \tag{17}$$

3. **Invariant projection** $\Pi_{\mathcal{C}_h}$: (a) normalize and clip; (b) truncate etendue; (c) zero OTF beyond D/λ; (d) enforce Strehl bound.

Key Advantage: Search Space Reduction

The unconstrained PSF lives in $\mathbb{R}_{\geq 0}^{N\times N}$ (typically $N = 256$ or 512, giving $\sim 10^5$ free parameters). The invariant constraints reduce the effective dimensionality dramatically: etendue bound reduces spatial–frequency support by 2×–10× per dimension; OTF cutoff eliminates frequencies above D/λ; symplectic structure (if enforced parametrically) reduces the PSF to 10–50 Zernike terms. The net effect: **invariant-constrained deconvolution searches a space that is 10^2–10^4 times smaller** than unconstrained blind deconvolution.

10.4.6 Identifiability Analysis via Fisher Information

A key weakness of standard blind deconvolution is the identifiability problem: the factorization $g = h * f$ is not unique. Any invertible filter ψ yields the alternative factorization $g = (h * \psi) * (\psi^{-1} * f)$, where both factors are valid solutions. This degeneracy is why blind deconvolution is ill-posed.

How invariant constraints resolve the degeneracy. The admissible set $\mathcal{C}_h$ eliminates most of the ambiguity because $h * \psi \in \mathcal{C}_h$ is far more restrictive than $h \in \mathcal{C}_h$:

1. **OTF cutoff (C2):** If h has OTF support $\leq f_c$ and ψ has support $\leq f_c$, then $h * \psi$ has support $\leq 2f_c$, violating the cutoff constraint. Thus only ψ with support ≤ 0 (i.e., $\psi = \delta$) preserves C2. This is the strongest resolvent of the degeneracy.
2. **Etendue bound (C1):** Convolution with ψ broadens the phase-space support, potentially violating the etendue constraint.
3. **Strehl bound (C5):** If $h(0) \leq h_{\text{Airy}}(0)$ and $\psi(0) \neq 1$, the product $h * \psi$ may violate the Strehl constraint.

Fisher information matrix. To quantify the improvement, we compute the Fisher information matrix (FIM) for the joint estimation of PSF parameters $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$ (Zernike coefficients) and scene parameters $\mathbf{f}$. For Gaussian noise with variance σ^2, the FIM elements are:

$$[\mathbf{F}]_{ij} = \frac{1}{\sigma^2} \sum_{x,y} \frac{\partial[h(\boldsymbol{\alpha}) * f](x,y)}{\partial \alpha_i} \cdot \frac{\partial[h(\boldsymbol{\alpha}) * f](x,y)}{\partial \alpha_j}. \tag{18}$$

In the unconstrained case, $\boldsymbol{\alpha}$ has N^2 components (one per PSF pixel), and the FIM is severely rank-deficient — the null space corresponds to the ambiguity directions ψ.

In the invariant-constrained case, $\boldsymbol{\alpha}$ has K components (Zernike coefficients, typically $K =$ 10–50), and the FIM is full-rank. The condition number improves from $\kappa_{\text{unconstrained}} \sim 10^8$–$10^{12}$ to $\kappa_{\text{constrained}} \sim 10^2$–$10^4$.

Practical consequence. The CRB for each Zernike coefficient is:

$$\text{Var}(\hat{\alpha}_i) \geq [\mathbf{F}^{-1}]_{ii}. \tag{19}$$

For the constrained problem with $K = 15$ Zernike terms and $N = 10^4$ photons per pixel, the CRB on coma (α_7) is approximately 0.02λ — well within the precision needed for PSF recovery. For the unconstrained problem, the FIM is singular and the CRB is formally infinite, confirming non-identifiability.

This analysis validates the claim that invariant constraints transform blind deconvolution from an ill-posed to a well-posed problem, with quantifiable estimation precision.

10.4.7 Connection to the Invariant Tower

Table 13: Invariant tower levels mapped to blind deconvolution.

Level	Invariant	Deconvolution Constraint
1	Fermat / OPL	OPD determines aberration structure
2	Lagrange H	Fixes PSF scale vs. magnification and NA
3	Etendue G	Bounds PSF phase-space support
4	Topological	PSF must carry correct topological charge
5	Fisher / CRB	Ultimate precision of deconvolved estimates

10.4.8 Practical Workflow: Invariant-Aware Deconvolution

1. **Invariant audit:** determine G, λ, D, H, approximate aberration class.
2. **Construct** $\mathcal{C}_h$.
3. **Run projected alternating minimization**.
4. **Validate** recovered h against all constraints.
5. **Assess**: if quality hits the CRB floor, need better data, not a better algorithm.

Central Finding

Blind deconvolution and forward optimization are two faces of the same invariant coin. The invariant-aware workflow applies identically: **audit first, compute second.**

11 Conclusion

This report establishes a unified framework for invariants in optical engineering, spanning forward design and inverse problems. The key contributions are:

1. **Invariant hierarchy:** Five levels from Fermat to Fisher, organized by symmetry and connected by the symmetry-breaking cascade (Section 2, Table 3).
2. **Symplectic bridge equation:** $\mathbf{M}^T\mathbf{J}\mathbf{M} = \mathbf{J}$ as the master identity for ray-optical invariants (Section 3).
3. **Information-geometric bridge:** The data processing inequality connects phase-space conservation (Levels 1–3) to Fisher information bounds (Level 5) via the space-bandwidth product (Section 3.5).
4. **Algorithmic invariants:** First catalog of eleven optimizer-proof quantities (Section 4, Table 5).
5. **Invariant tower:** Hierarchical structure across the ray-to-quantum continuum, with formal derivation via three mechanisms (Noether, DPI, symmetry breaking) and dual interpretation as hierarchy and layered constraint stack (Section 5.1, 5.2).
6. **Invariant-aware design:** Audit-first workflow with budget sheets, feasibility gates, and step-by-step procedure (Section 7).

7. **Three design rules:** Audit-before-optimize; design with margin; use topology at singularities (Section 8).
8. **FOM–algorithm mapping:** Operational connection between FOMs and algorithm classes for both forward optimization and inverse problems (Section 9.1, Table 11).
9. **Invariant-constrained blind deconvolution:** Conservation laws enforced as hard projections on the PSF, reducing search space by 10^2–10^4 and improving PSNR by 1–3 dB (Section 10.4, 6.3).
10. **Identifiability analysis:** Fisher information matrix confirms that invariant constraints transform blind deconvolution from ill-posed to well-posed, with quantifiable CRB on Zernike coefficients (Section 10.4.6).
11. **Worked numerical example:** Complete invariant audit for a Double Gauss machine-vision lens, demonstrating the workflow from specification to feasibility verdict (Section 6.4).

Central Theme

Invariants define the room in which all optical design takes place. Algorithms (B2) navigate the room. Singularities (B1) mark where the walls change shape. But it is invariants that tell you how big the room is — and whether your destination is inside it.
In the forward direction, invariants answer: *should I keep optimizing?* In the inverse direction, they answer: *should I keep deconvolving?* The answer in both cases is the same: **audit the invariants first, compute second.**

12 Looking Ahead: The Book (B3)

Table 14: From this report to the book: planned chapter map.

Part	Chapters	Scope
I. Foundations	Ch 1–4: Why Invariants Matter; Noether for Engineers; Lagrange Deep Dive; Etendue and the Second Law	Historical arc, rigorous derivations
II. The Tower	Ch 5–9: Abbe Sine Condition; Stokes Parameters; Topological Charge and Berry Phase; Quantum Fisher; SBP and the DPI Bridge	Each tower level with worked examples
III. Algorithmic	Ch 10–13: Definition and Catalog; Mode Count and Channel Capacity; Kramers-Kronig; κ_{TO}	Eleven algorithmic invariants
IV. Methodology	Ch 14–16: Invariant-Aware Workflow; Budget Sheets; Invariant-Based Tolerancing	Practical tools for industry
V. Inverse Problems	Ch 17–18: FOM–Algorithm Mapping; Invariant-Constrained Blind Deconvolution; Identifiability Analysis	ADMM formulation, worked examples
VI. Frontiers	Ch 19–21: AI and Invariant-Regularized Inverse Design; Freeform Invariants; Quantum Sensing Limits	Open problems, research directions

- **(B1) Optics after continuity:** Discontinuous Phase and the New Invariants Behind Singular Optics
- **(B2) Algorithms for optical engineering:** From Classical, Quantum-Inspired, to True Quantum
- **(B3) Invariants in optical engineering:** From Classical Laws to Physics-aware Optimization and Blind Deconvolution

References

[1] R. Welford, *Aberrations of Optical Systems* (Adam Hilger, 1986).

[2] W. T. Welford and R. Winston, *High Collection Nonimaging Optics* (Academic Press, 1989).

[3] M. Born and E. Wolf, *Principles of Optics*, 7th ed. (Cambridge University Press, 1999).

[4] R. R. Shannon, *The Art and Science of Optical Design* (Cambridge University Press, 1997).

[5] H. A. Buchdahl, *An Introduction to Hamiltonian Optics* (Dover Publications, 1993).

[6] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proc. R. Soc. Lond. A* **392**, 45–57 (1984).

[7] S. L. Braunstein and C. M. Caves, "Statistical distance and the geometry of quantum states," *Phys. Rev. Lett.* **72**, 3439 (1994).

[8] V. Giovannetti, S. Lloyd, and L. Maccone, "Quantum metrology," *Phys. Rev. Lett.* **96**, 010401 (2006).

[9] J. F. Nye and M. V. Berry, "Dislocations in wave trains," *Proc. R. Soc. Lond. A* **336**, 165–190 (1974).

[10] J.-L. Chern, "Optics After Continuity: Discontinuous Phase and the New Invariants Behind Singular Optics," Technical Report v35, March 2026.

[11] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed. (Johns Hopkins University Press, 2013).

[12] E. Noether, "Invariante Variationsprobleme," *Nachr. d. König. Gesellsch. d. Wiss. zu Göttingen*, 235–257 (1918).

[13] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed. (Springer, 1989).

[14] J. W. Goodman, *Introduction to Fourier Optics*, 4th ed. (W. H. Freeman, 2017).

[15] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.* **27**, 379–423 (1948).

[16] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006).

[17] R. A. Fisher, "On the mathematical foundations of theoretical statistics," *Philos. Trans. R. Soc. A* **222**, 309–368 (1922).

[18] A. W. Lohmann, "Space-bandwidth product of optical signals and systems," *J. Opt. Soc. Am. A* **13**(3), 470–473 (1996).

[19] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed. (Springer, 1999).

[20] S. Amari and H. Nagaoka, *Methods of Information Geometry* (AMS/Oxford University Press, 2000).

Symbol Glossary

Symbol	**Meaning**	**First Eq.**
$\mathbf{M}$	Ray-transfer matrix (ABCD)	(1)
$\mathbf{J}$	Symplectic matrix	(2)
H	Lagrange invariant	(3)
G	Etendue (phase-space volume)	(4)
$F(\theta)$	Fisher information	(5)
F_Q	Quantum Fisher information	(10)
N_{SBP}	Space-bandwidth product	(9)
ℓ	Topological charge	Table 3
$\mathcal{C}_h$	Admissible PSF set	(15)
$\Pi_{\mathcal{C}_h}$	Invariant projection operator	(17)
κ_{TO}	Tensor optical complexity	Table 5
Φ_{aug}	Augmented merit function	(11)
$\boldsymbol{\alpha}$	Zernike aberration coefficients	(18)
$\mathbf{F}$	Fisher information matrix	(18)